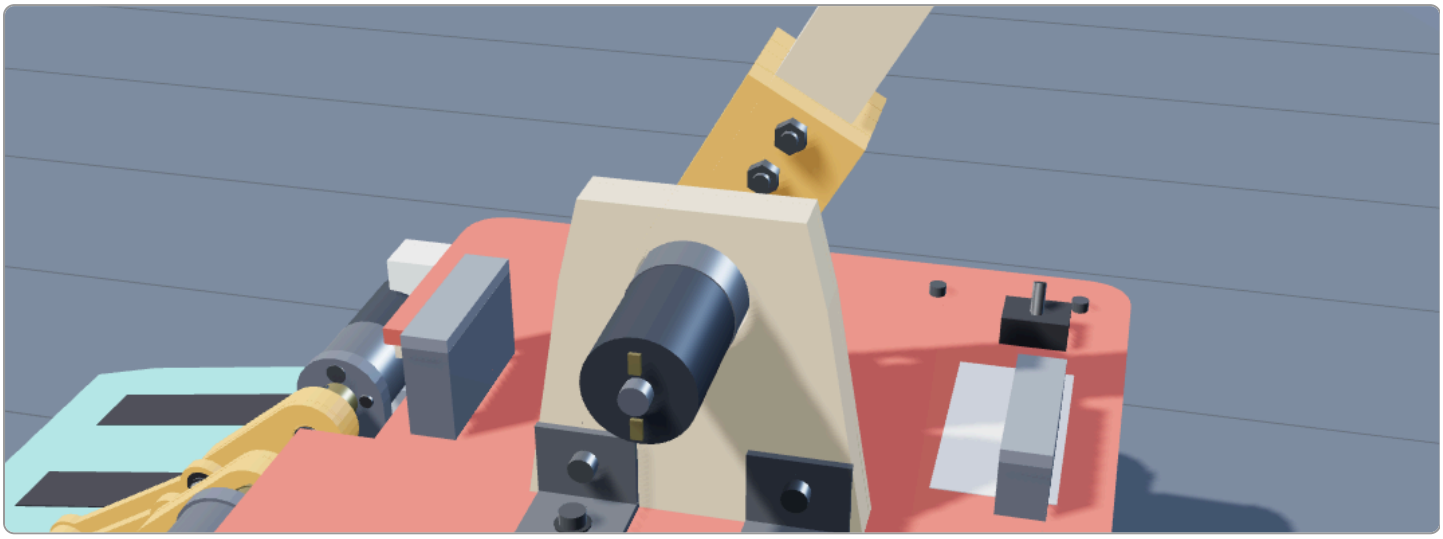
Wooden parts for a robot hammer

Hello! Please make the **4 wooden parts** below. Each part has its own page with a drawing at **real size** and simple steps.



This is how it will look on the robot: the light wood plate holds a motor, the stick and head swing over the top.

What to make

1. Motor plate
1 piece · 12 mm plywood
80 wide × 78 tall
page 2–3

2. Stick
1 piece · hard wood (beech / oak)
20 × 15, 222.5 long
page 4

3. Two small blocks
2 pieces · any wood
44 × 12, 14.5 and 25 tall
page 4

4. Hammer head
1 · hard wood, glued from 4 pieces
64 × 30 × 25
page 5

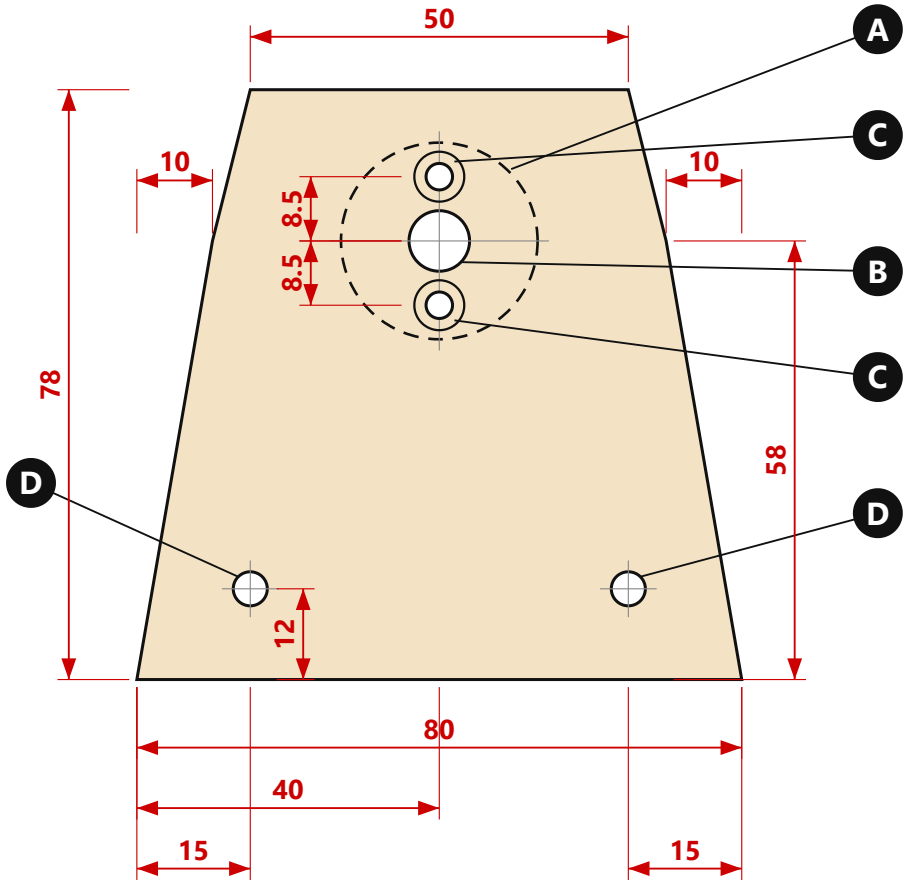
How to read the drawings

All numbers	are **millimetres (mm)**. 10 mm = 1 cm.
Red arrows	a measurement: the number is the distance between the two arrow tips.
A **B** …	a hole. The table under the drawing says which **drill bit** and **how deep**.
- - - dashed line	something on the **other side** of the wood (you can't see it from this side).
Real size	print the pages at **100 %** ("actual size", not "fit to page"). Then the drawing is exactly as big as the part – you can cut it out, glue it on the wood and drill through the marks. Check: the black-and-white bar must measure **50 mm** with a ruler.

Please do not drill holes in the stick or the head – we drill those ourselves when we put everything together, so they line up.

1 · Motor plate – drawing (real size)

1 piece of **12 mm plywood**. This is the **FRONT side**. The motor goes on the **BACK side** (the dashed big circle is on the back).



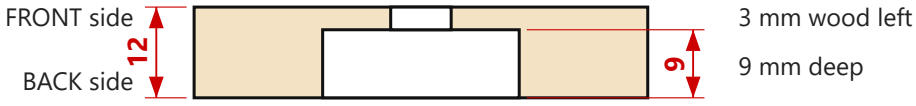
the plate is the same both ways round – only FRONT / BACK side matter

this bar must measure 50 mm

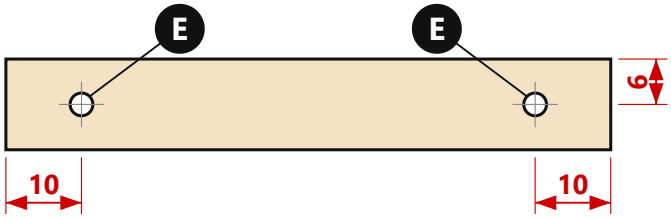
Tip: cut this page out along the outline, glue it on the plywood, cut the shape, then drill through the centre of each circle.

1 · Motor plate – holes and steps

Hole	How many	Drill bit	How deep	Where
A	1	**26 mm flat-bottom bit (Forstner)**	**9 mm only – do NOT go through**	from the **BACK** side, in the middle (40 from each side), 58 up from the bottom
B	1	**8 mm**	all the way through	in the middle of hole A
C	2	**3.5 mm**	through	8.5 above and 8.5 below the middle of A. On the **FRONT** side widen the top a little (countersink) so a flat screw head sits flush
D	2	**4.5 mm**	through	12 up from the bottom, 15 in from each end
E	2	**3 mm**	25 mm deep	into the **bottom edge**, 10 in from each end, in the middle of the 12 mm thickness



hole A seen from above, cut in half: 9 deep, 3 mm wood stays

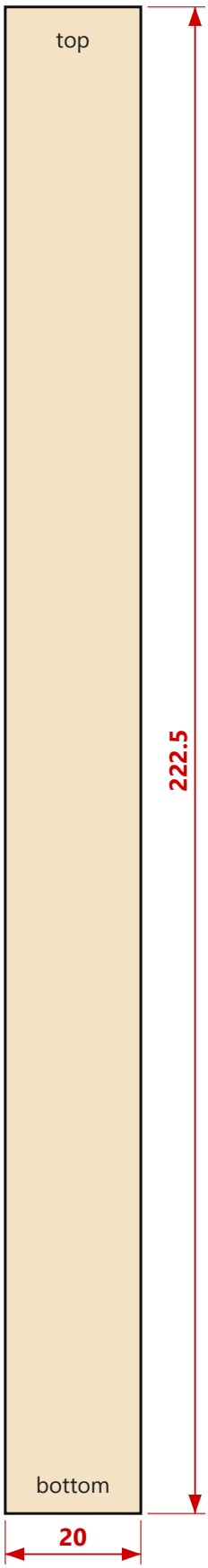


the bottom edge (the 12 mm side that stands on the robot)

Steps

1. Cut the shape from 12 mm plywood: 80 wide at the bottom, 50 wide at the top, 78 tall (see page 2). Sand the edges.
2. Mark the holes (or glue the real-size page 2 on the wood).
3. Turn the plate over to the **BACK** side. Drill hole **A** with the 26 mm flat-bottom bit, **only 9 mm deep** – stop when 3 mm of wood is left (tape on the drill bit at 9 mm helps).
4. Drill **B** (8 mm) through the middle of A.
5. Drill the 2 holes **C** (3.5 mm) through. On the **FRONT** side, widen their top with a countersink bit (or a 7 mm bit, just 2 mm).
6. Drill the 2 holes **D** (4.5 mm) through.
7. Stand the plate on its bottom edge and drill the 2 holes **E** (3 mm) straight down, 25 mm deep.

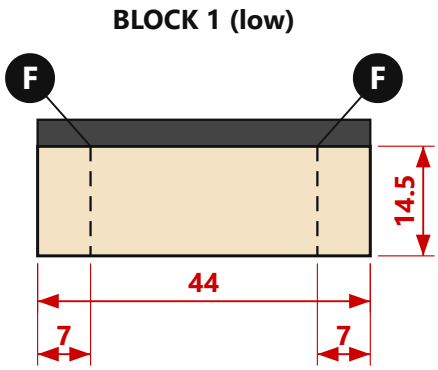
2 · Stick and 3 · Two small blocks



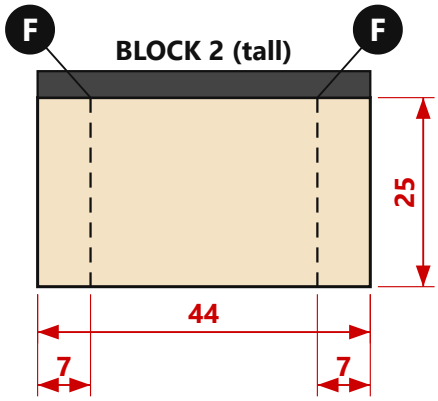
2 · Stick – 1 piece

1. Hard wood (beech or oak), **straight grain, no knots**.
2. Cut a strip **20 mm × 15 mm**.
   Exactly 20 × 15, or a tiny bit smaller (19.8 × 14.8) – **not bigger**: the top end must slide into a plastic part with a 20 × 15 hole.
3. Cut it to **222.5 mm** long, square ends.
4. **No holes** – we drill them ourselves.

3 · Two small blocks – 2 pieces



12 mm thick · rubber 3.5 mm on top



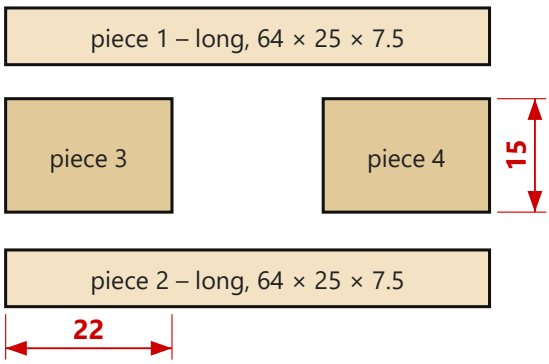
12 mm thick · rubber 3.5 mm on top

1. Cut 2 blocks, both **44 long × 12 thick**: block 1 is **14.5** tall, block 2 is **25** tall.
2. In each: 2 holes **F** **3.5 mm**, from the top straight through to the bottom, 7 mm from each end, in the middle of the 12 mm.
3. Glue a piece of 3.5 mm rubber (44 × 12) on top.

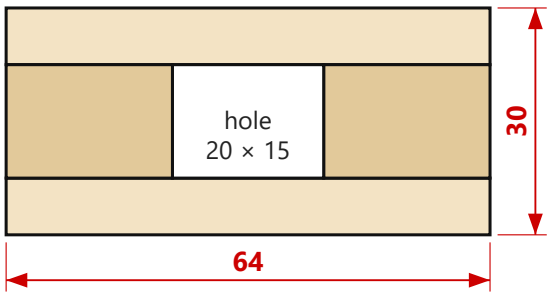
4 · Hammer head – made from 4 glued pieces

Gluing 4 simple pieces is easier than cutting a square hole into one block. Use **hard wood**; all pieces are **25 mm tall**.

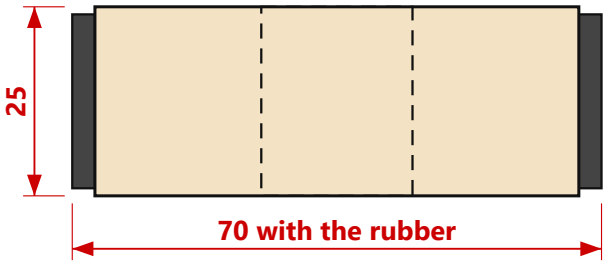
THE 4 PIECES (seen from above, full size)



GLUED TOGETHER (from above)



FROM THE SIDE: 25 mm tall, rubber on both ends



rubber pad 28 × 23 × 3,
glue one on each end



this bar must measure 50 mm

Piece	How many	Size (long × tall × thick)
1 and 2 – long pieces	2	**64 × 25 × 7.5**
3 and 4 – small blocks	2	**22 × 25 × 15**
Rubber pads	2	28 × 23, 3 mm thick

Steps

1. Cut the 4 pieces. All of them exactly **25 mm** tall.
2. Glue: long piece – 2 small blocks at the two ends – long piece (like a sandwich, see the drawing). The gap in the middle must be a hole **20 × 15** going all the way through, top to bottom. Put the stick in the hole while the glue dries so it stays the right size (do not glue the stick!).
3. Clamp and let it dry. Sand the outside flat: 64 × 30 × 25.
4. Glue a rubber pad on each 30 × 25 end.
5. **No other holes** – we drill the bolt hole ourselves.

Finished? Please check

☐ Motor plate: 80 wide, 78 tall, 12 thick. Hole **A** is only 9 mm deep (3 mm of wood left) and on the **back** side.

☐ Motor plate: holes **C** are countersunk on the **front** side (the side without hole A).

☐ Motor plate: 2 small holes **E** in the bottom edge.

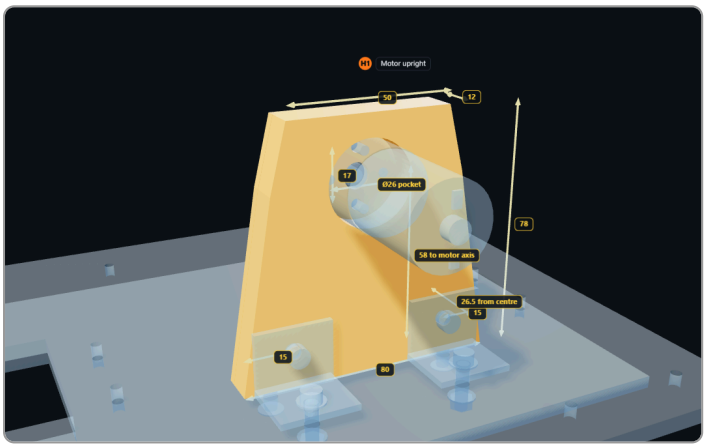
☐ Stick: 20 × 15 (not bigger), 222.5 long, no holes.

☐ Stick slides into the hole of the head without forcing.

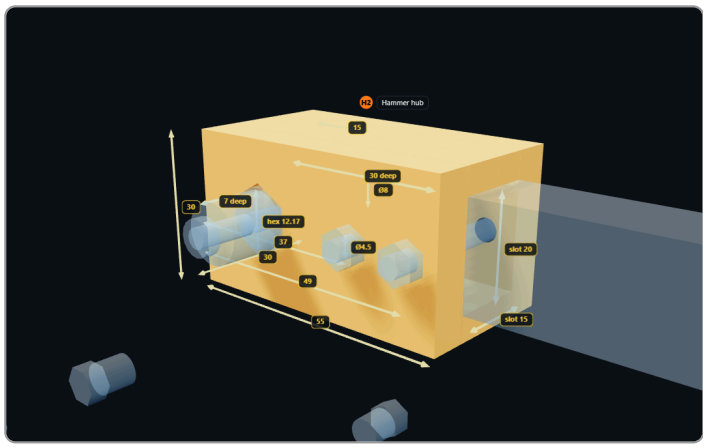
☐ Head: 64 × 30 × 25, rubber on both ends, 20 × 15 hole through the middle.

☐ 2 blocks: 14.5 and 25 tall, 2 holes each, rubber on top.

How we will use the parts



The motor (blue) sits in the 26 mm hole of the plate; 2 metal angles bolt the plate to the robot.



A 3D-printed part (orange) goes on the motor; the stick (blue) slides into it and we bolt it.

Thank you! If something is not clear, call us before cutting – it is easier to ask than to make it again.